SC4001 · PROJECT E

Catching an AI's wrong answers about where things are

Goal: Build a small lie detector that spots, from the AI's hidden layer activations, when it gets "left of", "on top of" and similar relations wrong.

EXAMPLE

Photo: a **cup to the left of a plate**.

We ask: "Is the cup to the right of the plate?" The AI says: **Yes**

green what the photo shows red a wrong answer: the detector should catch it

Proposed process

1 Get the data

Use the Visual Genome dataset - photos where people boxed the objects and wrote how they relate.

`objects file` `relationships file` `alias files`

`out → photos + (object, relation, object) lists`

2 Keep only position relations

We keep 8 relations. The alias file groups different wordings ("left of", "to the left of") into one.

`left · right` `up · down` `front · back` `on · under`

`out → clean list of relations per photo`

3 Turn each relation into questions

Each relation gives several questions with a known answer. Photos are then split into a train set and a test set.

ANSWER: YES

True version

1. "Is the cup to the left of the plate?"
2. The same question in other words.

ANSWER: NO

Flipped version

1. "Is the cup to the right of the plate?"
2. "Is the plate to the left of the cup?"

`out → questions + ground truth, split by photo`

4 Ask the AI each question

For each one we save its answer, how confident it was, and its hidden layer activations.

`Qwen 3B · frozen, not trained`

`out → answer + confidence + internal numbers for each question`

5 Find the wrong answers

We compare each answer with the ground truth.

`out → each answer labelled right or wrong`

6 Train the lie detector

A small model learns to tell wrong answers from right ones using the hidden layer values. Each of us tries a different detector design and we keep the best.

`only the detector is trained`

`out → detector that says right or wrong`

7 Test on new photos

We use test-set photos the detector never saw. Is it better than the AI's own confidence?

`AUROC · F1` `vs answer confidence` `per relation`

Coordinate on

DATASET
Visual Genome
Same cleaned relations and same train / test split for everyone.

MODEL
Qwen 3B
Same checkpoint, precision, image size and prompt.

DATA SHAPE
One record per question
Same fields and same hidden-state layout, so detectors can be compared.

Runs on a free Google Colab T4 GPU. Data: Visual Genome. Model: Qwen 3B (Qwen2.5-VL-3B-Instruct, to confirm).